

ORCA: Live Interactive Demo Walkthrough & Screen Narration Script

Choreographed 6-Minute Live System Pitch for the Smart India Hackathon Grand Jury

Problem Statement ID: SIH26176	Team Name: CodeCatalysts
Demo Target Duration: 6 to 7 Minutes Strict	System URL: http://127.0.0.1:8000
Live Stack: FastAPI + Leaflet + 60 FPS Windy Canvas	Presenter Setup: Dual screen (Laptop + Projector)

■■ LIVE DEMO PRE-FLIGHT CHECKLIST (COMPLETE 10 MINS BEFORE JURY ARRIVES)

1. Start FastAPI server: `python -m uvicorn backend.main:app --port 8000`
2. Open Chrome browser to <http://127.0.0.1:8000> in Full-Screen mode (Press F11).
3. Verify Leaflet map tiles render and live coordinates badge tracks cursor (█ 9.9312°N, 76.2673°E).
4. Ensure microphone permissions are allowed for Web Speech API voice demonstrations.
5. Keep sample prompt chips ready in chat feed so zero typing is required if under tight time pressure.

ACT 1: THE OCEAN COCKPIT & LIVE CARTOGRAPHY

■■ Target Duration: 0:00 – 0:50 (50 Seconds)

On Screen Action: Presenter opens browser at <http://127.0.0.1:8000> in Full-Screen mode. The screen shows the Ocean Cockpit with a dark tactical Leaflet nautical chart of coastal India on the left, and the Docked Copilot Chat on the right. Move cursor over Kochi and Mumbai to show the live header coordinate chip updating in real time.

■■ Spoken Narration (What You Say to the Jury):

"Respected jury, welcome to ORCA. What you see on screen is not a static Figma mockup or a generic chat wrapper — this is our live, operational single-page marine intelligence cockpit.

On the left is our enterprise cartographic engine displaying India's Exclusive Economic Zone (EEZ), the International Maritime Boundary Line (IMBL), official Marine Protected Areas like Gahirmatha and Gulf of Mannar, along with 35+ pre-indexed coastal harbors and NIOT OMNI data buoys.

Notice the live coordinate telemetry badge at the top tracking my cursor with sub-meter geodetic precision. Now, let us put our 8 Collaborative Agents to the test with a real-world fisherman's inquiry."

■ Jury Psychological Impact: Judges immediately see a functional, beautiful, real-time tactical marine GIS system.

ACT 2: MULTI-AGENT COPILOT & ZERO-HALLUCINATION TRACE

■■ Target Duration: 0:50 – 2:05 (75 Seconds)

On Screen Action: Click the microphone icon or click the suggested chip: 'Is it safe to fish near Kochi tomorrow morning and what are the sea conditions?' Watch the response generate. Instantly click on the '**Agent Execution Trace**' accordion to expand the step-by-step agent timings (Planning Agent -> Marine Retrieval -> Weather Intel -> Risk Agent -> Synthesis).

■■ Spoken Narration (What You Say to the Jury):

"I will now ask ORCA: 'Is it safe to fish near Kochi tomorrow morning and what are the sea conditions?'"

In less than 1.2 seconds, look at what happens behind the scenes in our **Agent Execution Trace**:

- **Planning Agent** analyzed the natural language, resolved Kochi to latitude 9.9656°N, longitude 76.2425°E, and selected the domain specialists.
- **Marine Data Retrieval Agent** pulled real-time wave heights (1.2m) and swell from Open-Meteo and in-situ buoys.
- **Weather Intelligence Agent** calculated Douglas Sea State 2 and Beaufort Wind Force 3.
- **Risk Assessment Agent** executed our deterministic safety matrix: Risk Score 16 out of 100 — giving a clear, unambiguous **GREEN: SAFE TO VENTURE** verdict.

Notice this crucial guarantee: **The LLM did not guess these numbers.** If I expand the **Data Provenance Ledger** at the bottom, you can see the exact satellite pass, dataset ID (ECMWF Wave 0.25°), observation timestamp, and retrieval latency. Everything is verifiable."

■ **Jury Psychological Impact: Demonstrates the 8 agents in action, zero hallucination, and full data provenance auditability.**

ACT 3: 60 FPS WINDY-STYLE FLUID SIMULATOR & UNIT SWITCHER

■■ Target Duration: 2:05 – 3:15 (70 Seconds)

On Screen Action: Click on the right-hand map toolrail. Toggle '**Wind Flow**'. Watch dynamic colored particle streamlines glide smoothly over the Indian Ocean. Then toggle '**Swell Waves**' to show undulating crest ripple lines. Then toggle '**Sea Surface Temperature**' to show the vibrant thermal gradient heatmap. Finally, click the bottom scale bar unit badge (^ `kt` -> `km/h` -> `m/s`).

■■ Spoken Narration (What You Say to the Jury):

"Now let us look at our physical ocean fluid engine. Most hackathon apps paste static opaque heatmaps that obscure the nautical chart. In ORCA, we engineered a custom **HTML5 2D Canvas engine running locked at 60 FPS** directly synchronized over Leaflet.

As I toggle **Wind Flow**, you see global velocity streamlines with speed-colored motion blur.

Switching to **Swell Waves**, you see authentic Windy-style undulating wavefront ripple lines that immediately tell a vessel master the direction of rough oceanic breakers.

Switching to **Sea Surface Temperature**, our satellite radiometry layer renders the exact thermal fronts where cold nutrient-rich upwelling meets warm surface waters.

And observe this: Artisanal fishermen in Kerala use knots, while Gujarat fishermen prefer km/h, and navy masters use meters per second. With our **1-Click Unit Switcher** on the bottom scale bar, the entire application, legend, and cursor probe instantly convert units without re-fetching data."

■ Jury Psychological Impact: Wows the judges with fluid physics graphics and fisherman-centric 1-click UX.

ACT 4: SAFE MARINE ROUTE ENGINE & VOYAGE SIMULATOR

■■ Target Duration: 3:15 – 4:30 (75 Seconds)

***On Screen Action:** Navigate to the **Routes** tab. Select Origin: Cochin Port (Kochi) and Destination: Port of Colombo (Sri Lanka). Click '**Calculate Safe Marine Routes**'. Two polyline paths render on the map (Route Alpha in Blue, Route Bravo in Green). Scroll to the steerage table. Click '**Start Voyage Simulator**'. Watch a vessel icon glide along waypoints.*

■■ Spoken Narration (What You Say to the Jury):

"Now, let us examine our marine navigation engine — one of the most technically challenging aspects of SIH26176.

I have set a voyage from Cochin Port to Colombo. Look at the two calculated paths:

- **Route Alpha** follows the coastal fairway (185 NM).
- **Route Bravo** is our deep-water bypass route (208 NM), which arcs seaward to avoid shallow reefs and rough inshore chop, boosting safety from 82 to 94.

Look at the southern tip of India: Notice how the route curves smoothly around Cape Comorin Rounding (7.65°N, 77.40°E) and deep into the Wadge Bank fairway. **There is zero land crossing.** Our fairway algorithm strictly avoids the peninsular landmass.

Below, our naval resistance power curve computes fuel burn: saving over 200 liters of diesel. And as I click '**Start Voyage Simulator**', our live vessel simulator animates the ship along each waypoint, displaying turn-by-turn compass steerage like Steer 142° SE with Douglas sea state at every coordinate."

■ Jury Psychological Impact: Proves 0% land crossing, Cape Comorin rounding, fuel ROI, and live voyage simulation.

ACT 5: POTENTIAL FISHING ZONE (PFZ) HOTSPOTS

■ Target Duration: 4:30 – 5:20 (50 Seconds)

On Screen Action: Switch to the **PFZ** tab. Show the grid of active fishing zones around Kochi and Mangalore. Toggle the fleet filter from 'Artisanal Motorized (OBM, <15 NM)' to 'Deep-Sea Commercial Trawler (15-120 NM)'. Click '**Plot Safe Route**' on the top Yellowfin Tuna card.

■ Spoken Narration (What You Say to the Jury):

"On our **Potential Fishing Zones** tab, ORCA translates satellite oceanography into direct fishermen income.

Instead of guessing, ORCA scans thermal fronts where temperature gradient is $\Delta T \geq 0.5^{\circ}\text{C/km}$ and chlorophyll-a exceeds 1.0 mg/m^3 , indicating rich phytoplankton blooms.

Crucially, we don't send small artisanal canoes 40 nautical miles out into dangerous deep waters. Our **Fleet Profile Filter** restricts small craft to safe nearshore sectors ($\leq 15 \text{ NM}$), while unlocking deep pelagic tuna corridors for commercial trawlers.

Each card shows target species (Yellowfin Tuna, Mackerel), recommended gear (Ring seine / Gillnet), compass bearing, depth soundings, and estimated fuel burn. With one click on '**Plot Safe Route**', the waypoint is sent directly to the map stage for immediate departure."

■ **Jury Psychological Impact:** Directly links satellite thermal fronts with fisherman economic prosperity and safety.

ACT 6: EMERGENCY GMDSS MAYDAY SOS BEACON & PDF RECEIPT

■ Target Duration: 5:20 – 6:15 (55 Seconds)

On Screen Action: Click the pulsing Red '**EMERGENCY SOS**' button in the top header. In the modal, select 'Engine Failure & Drifting Offshore' with 4 crew members. Click '**TRANSMIT MAYDAY BEACON**'. Show the response dispatching to MRCC Kochi and identifying patrol ship ICGS Samar. Click '**Download PDF SOS Receipt**'.

■ Spoken Narration (What You Say to the Jury):

"Finally, what happens when disaster strikes at sea? Every year, dozens of Indian fishermen lose their lives due to delayed distress calls.

With one tap on ORCA's **GMDSS Emergency SOS Beacon**, the system automatically detects our coordinates, calculates that the nearest Coast Guard station is **MRCC Kochi**, computes a **Search and Rescue response ETA of 24 minutes**, and scans coastal AIS traffic to identify that the Coast Guard patrol vessel **ICGS Samar** is just 3.5 nautical miles away monitoring VHF Channel 16.

It instantly generates the official phonetic voice Mayday script: MAYDAY, MAYDAY, MAYDAY. THIS IS IND-KL-07-ORCA... and issues a cryptographic SHA-256 dispatch token.

I can now click '**Download PDF Distress Receipt**', generating an official, legal document with coordinates and timestamp that can be shared with port authorities."

■ **Jury Psychological Impact:** High-impact emotional and technical climax showcasing real life-saving capabilities.

ACT 7: GRAND FINALE & INVITATION TO JURY Q&A;

■■ Target Duration: 6:15 – 6:45 (30 Seconds)

On Screen Action: Close modal, zoom out Leaflet map to show Pan-India coastline with live animated wind particles running smoothly in the background.

■■ Spoken Narration (What You Say to the Jury):

"To conclude: ORCA is not a toy, a mock-up, or an unverified chatbot. It is a robust, production-ready, multi-agent marine intelligence platform backed by 34 passing automated tests, grounded in verified government datasets, and engineered for India's Blue Economy.

We thank you for your time, and we are eager to answer your technical and architectural questions. Thank you!"

■ Jury Psychological Impact: Leave judges deeply impressed with working software and invite questions confidently.